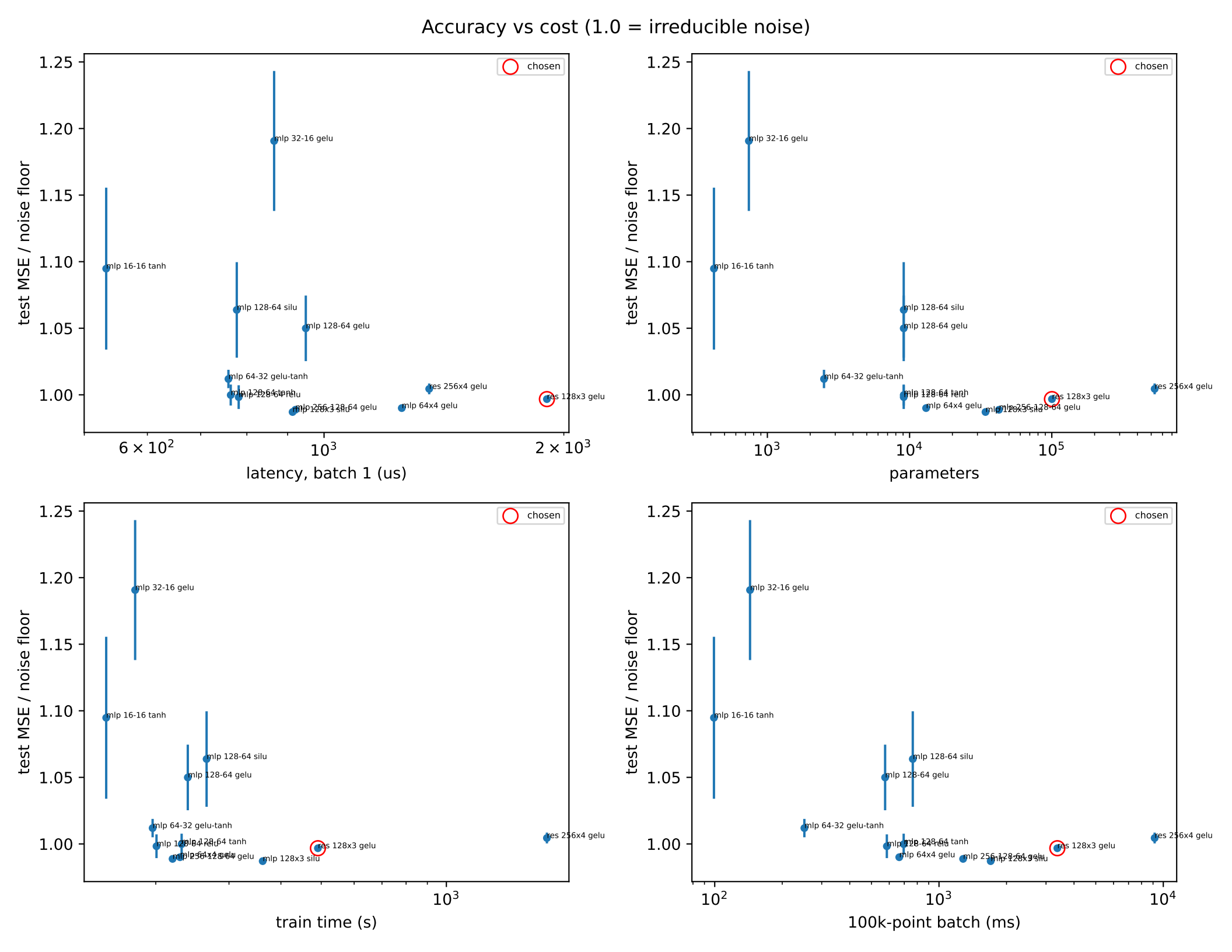
3D DNN SURROGATE - ARCHITECTURE SEARCH REPORT

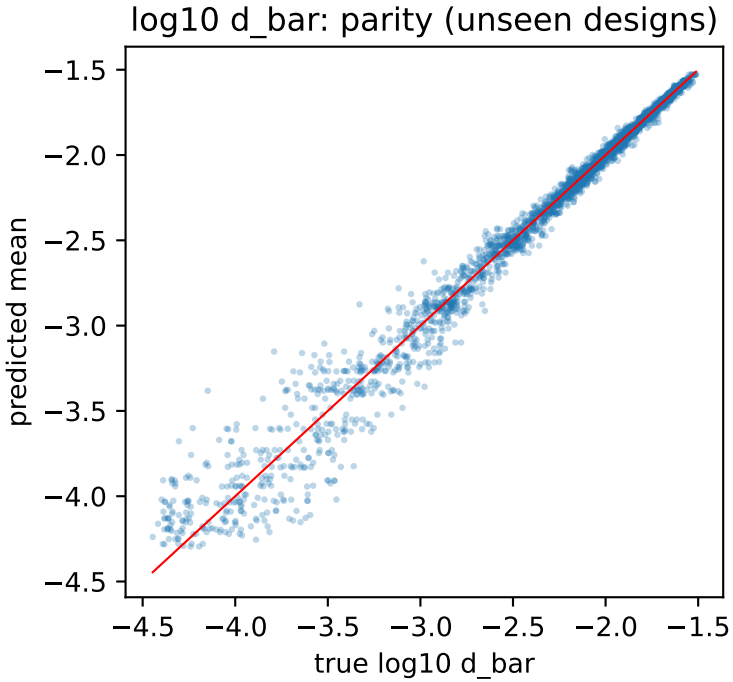
GOAL: replace the GP surrogate in ABC-MCMC. For theta=(log10 p1, log10 p2, tau) predict the
joint distribution of the two ABC features of the J=100 cultures: d_bar (mean of (X/Z)^(1/4)) and s (its sd),
fast and with an honest spread.   Data: 50,000 rows (10,000 LHS designs x 5 reps), paper box, tp=20, delta=1.
29.6% of rows have d_bar = 0, so the model is a hurdle model:
P(d_bar=0) (logistic head) + Gaussians on (log10 d_bar, log10 s) for nonzero (mean, log-variance heads).
Split BY DESIGN point 80/10/10 (unseen theta), 60 fits = 12 architectures x 5 seeds, 16 cores, 32.9 min.
Noise floor (irreducible within-design variance), test: log10 d_bar 8.07e-03, log10 s 6.86e-03.

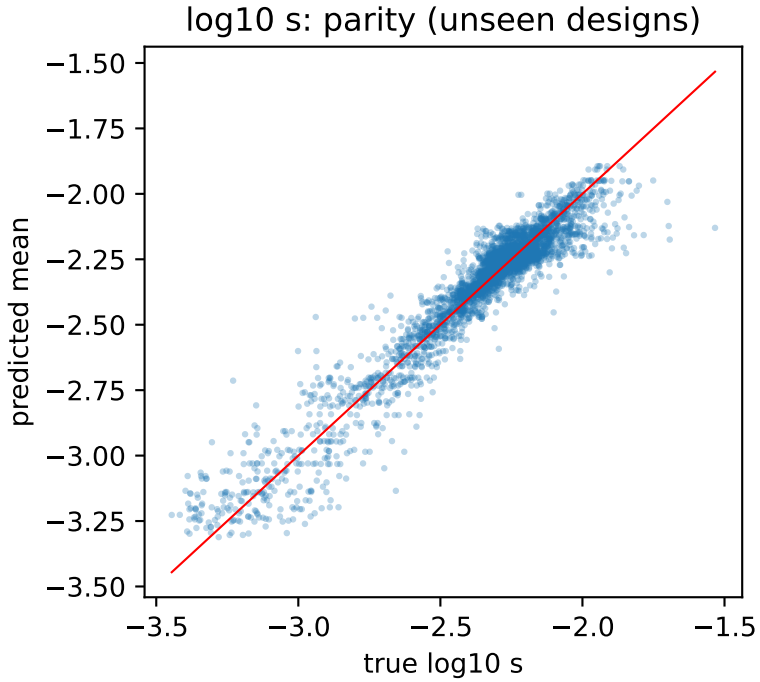
SELECTION (rule fixed in advance): best mean validation loss = res_128x3_gelu;
chosen = cheapest model within 1 SE (0.0017) of it = res_128x3_gelu
Chosen model (seed 0, saved to surrogate_3d.pt) on held-out TEST designs:
  nonzero part (d_bar | s): MSE = 0.99 | 1.00 x noise floor, RMSE 0.0895 | 0.0827 (log10),
                95% coverage 0.949 | 0.948 after conformal (raw 0.947 | 0.947)
  zero part   : log-loss 0.0458, Brier 0.0143, AUC 0.9985
GP baseline (GPS-ABC style, 300 pts, nonzero part only, cannot model the zero atom):
  log10 d_bar only: RMSE 0.1760 = 3.84 x floor, 95% coverage 0.885, fit 5s, query 18922 us

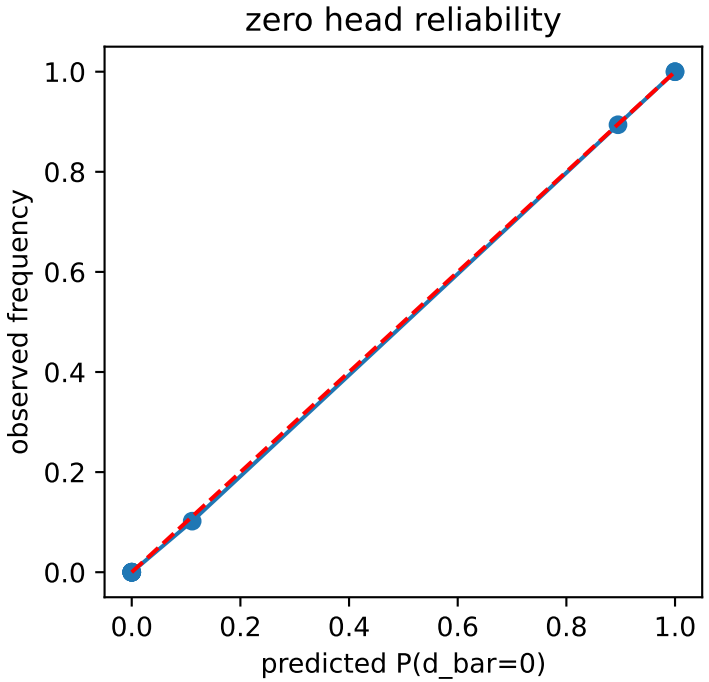
Mean over 5 seeds (sorted by validation loss; test_* are held-out designs; latency = batch 1, microseconds)

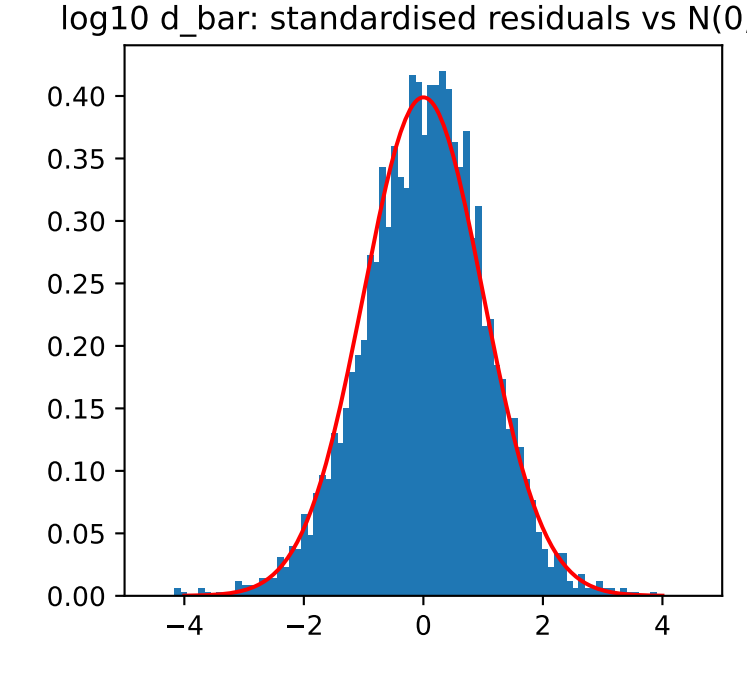
arch	params	val_loss	test_x_floor	test_rmse	test_nll	test_ll0	test_auc0	test_cov95_conf	latency_us	batch1e5_ms	train_s	x_floor_sd
res_128x3_gelu	100,229	-3.287	0.9968	0.0862	-3.145	0.04569	0.9985	0.947	1905	3370	490.9	0.001672
mlp_256-128-64_ge	42,501	-3.284	0.9888	0.08585	-3.147	0.04548	0.9985	0.948	920.4	1281	219.5	0.002326
mlp_64x4_gelu	13,061	-3.284	0.99	0.0859	-3.141	0.04522	0.9985	0.9472	1252	664.9	228.7	0.001865
mlp_128x3_silu	34,181	-3.279	0.9871	0.08578	-3.151	0.045	0.9985	0.9499	913	1700	361.6	0.002424
res_256x4_gelu	528,645	-3.27	1.004	0.08653	-3.118	0.04568	0.9985	0.9467	1355	9151	1745	0.004048
mlp_128-64_gelu	9,093	-3.254	1.05	0.08851	-3.112	0.0454	0.9985	0.9514	948.6	575.1	238.8	0.02463
mlp_128-64_relu	9,093	-3.253	0.9983	0.08626	-3.114	0.04548	0.9985	0.9497	781.3	585.4	200.8	0.008919
mlp_128-64_tanh	9,093	-3.249	0.9998	0.08633	-3.114	0.04552	0.9985	0.9528	763.5	695.9	230.8	0.007894
mlp_64-32_gelu-ta	2,501	-3.24	1.012	0.08685	-3.102	0.04509	0.9985	0.9525	758.5	251	196.5	0.006894
mlp_128-64_silu	9,093	-3.24	1.064	0.08909	-3.097	0.04518	0.9985	0.9531	777	764	265	0.03588
mlp_16-16_tanh	421	-3.155	1.095	0.09033	-3.009	0.04771	0.9984	0.9535	532.7	99.21	152	0.06082
mlp_32-16_gelu	741	-3.127	1.191	0.09426	-2.991	0.04809	0.9983	0.9527	865.6	143.7	178.4	0.05256

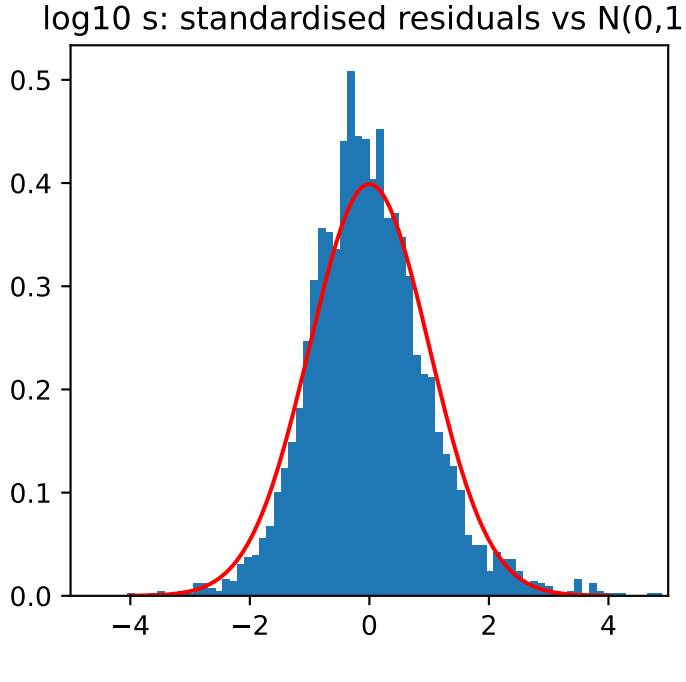

Accuracy vs cost (1.0 = irreducible noise)
test MSE / noise floor
latency, batch 1 (us)
parameters
train time (s)
100k-point batch (ms)
chosen
mlp 32-16 gelu
mlp 16-16 tanh
mlp 128-64 silu
mlp 128-64 gelu
mlp 64-32 gelu-tanh
mlp 128-64 tanh
mlp 128-64 relu
mlp 64x4 gelu
mlp 256-128-64 gelu
mlp 128x3 silu
res 256x4 gelu
res 128x3 gelu

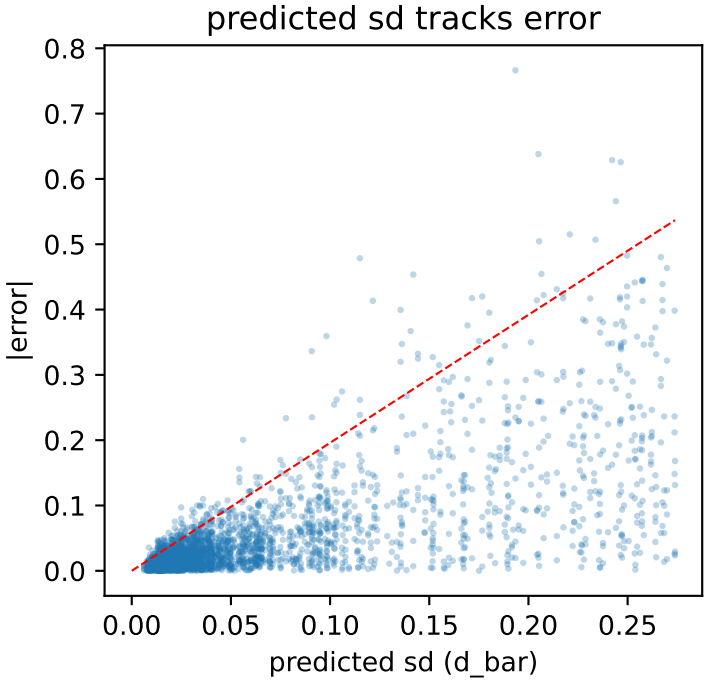

log10 d_bar: parity (unseen designs)
predicted mean
true log10 d_bar
−1.5
−2.0
−2.5
−3.0
−3.5
−4.0
−4.5

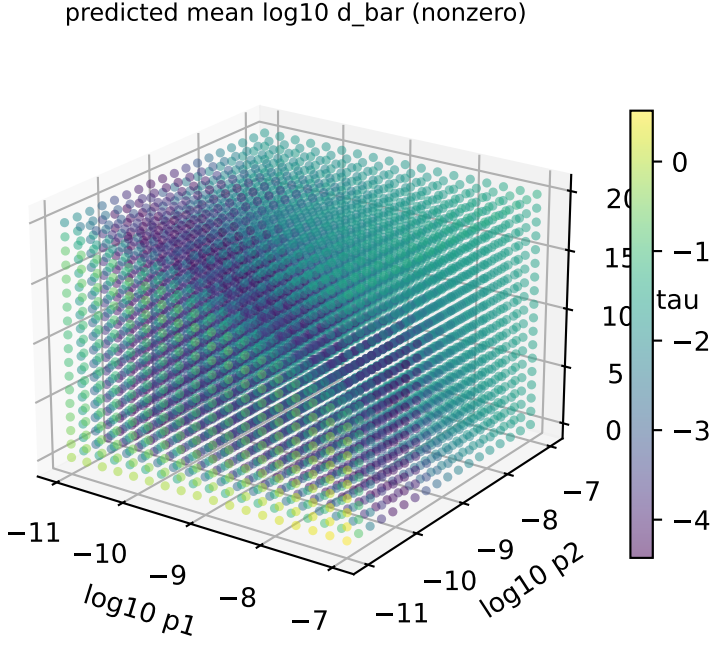

log10 s: parity (unseen designs)
predicted mean
true log10 s
−1.50
−1.75
−2.00
−2.25
−2.50
−2.75
−3.00
−3.25
−3.50
−3.5
−3.0
−2.5
−2.0
−1.5

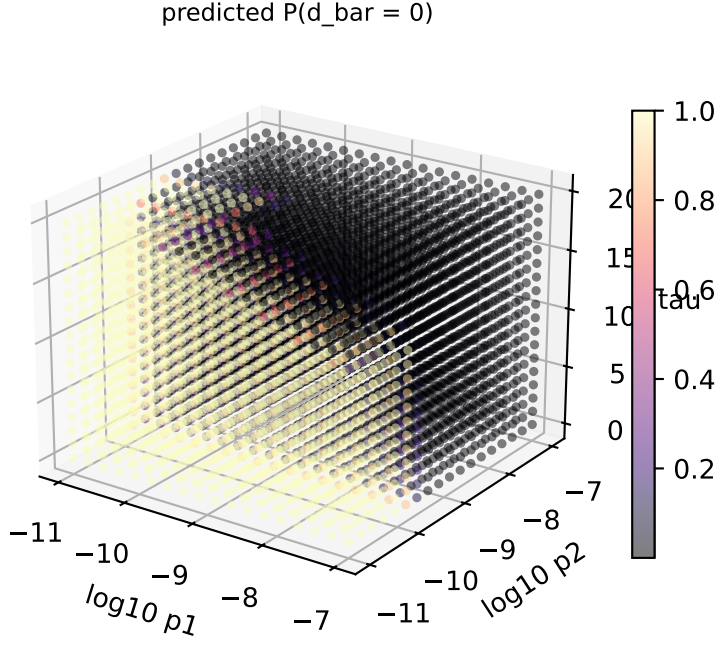

zero head reliability
observed frequency
predicted P(d_bar=0)
0.0
0.2
0.4
0.6
0.8
1.0


log10 d_bar: standardised residuals vs N(0,1)
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
−4
−2
0
2
4


log10 s: standardised residuals vs N(0,1)
0.0
0.1
0.2
0.3
0.4
0.5
−4
−2
0
2
4


predicted sd tracks error
|error|
predicted sd (d_bar)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.00
0.05
0.10
0.15
0.20
0.25


predicted mean log10 d_bar (nonzero)
log10 p1
log10 p2
tau


predicted P(d_bar = 0)
log10 p1
log10 p2
tau